Cross-Asset Signals for FX Futures Trading

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Strategy overview

- Use excess equity market returns to identify FX futures opportunities
- Generate signals by isolating FX movements related to country's equity performance
- … (More to come once performance analysis is complete)

Motivation

- Strong foreign equity performance attracts capital
- Investing in foreign markets requires buying and transacting in the local currency
- Problems:
  - **Exchanging Currencies is time consuming and inefficient because of how FX markets work**
  - **FX rate appreciates when investors rebalance in the same direction and the market cannot absorb order flow**
- Goal: Profit of the lead-lag relationship caused by such problems

Trading Hypothesis 1

- There exists a lead-lag relationship between a country's stock indices and the exchange rate of the country's currency
  - **Ex: A strong performance of the Nikkei 225 will be followed by an appreciation of the yen in FX markets**
- Economic Intuition: A strong performance in a country's equity markets will attract international investors looking to take advantage of a bull market. These investors will need to convert to the foreign currency, driving up the exchange rate

Trading Hypothesis 2

- The lag in the relationships previously mentioned, will be larger in less liquid FX markets
  - **Ex: The lag in the currency appreciation in Zimbabwe will be larger than the lag in Great Britain**
- Economic Intuition: Less liquid FX markets adjust more slowly to new information because trading depth is limited. As a result, capital flows take longer to move prices, leading to larger lags in exchange rate responses

Data

Data Sources

- CDS Data: Bloomberg
- Equity Data: Bloomberg
- FX Spot Rate Data: FRED
- FX Future Data: CME

Investment Universe

- FX Futures (CME)

Note: As proof of concept, investment universe for hypothesis testing is restricted to…

- Australia
- Brazil
- Canada
- Great Britain / United Kingdom
- Japan
- Mexico
- South Africa

Trade Implementation

- Step 1: Remove systematic macro and FX factors from excess FX returns
  - **FX Factors: Hedge out FX Carry and US Dollar Strength**
  - **Macro Factors: Hedge out macro factors using CDS spread as a proxy**
  - **Hedging out these factors isolates currency-specific movements**
- Step 2: Generate Trading Signal
  - **Define excess equity returns as (Foreign index - S&P 500)**
  - **Regress isolated currency movement on excess equity returns**
  - **Use the regression above to generate trading signal**

Signal Evaluation

- Signal generation occurs at end of day **t**
- Gain exposure to our signal via CME FX futures rather than FX rates because
  - **it is easier to capture realistic trading costs**
  - **futures have consistent roll over schedules**
  - **futures have high liquidity during US trading hours**
- Trade execution occurs at start of day **t+1**
  - **Position size is determined by signal sign and magnitude**

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